

Mach-Zehnder Interferometers for Group-Index Extraction

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Abstract. Five unbalanced Y-branch Mach-Zehnder interferometers (MZIs) with path-length differences $\Delta L = 51\text{-}501\text{ }\mu\text{m}$, and a grating-coupler loopback, are designed in the SiEPIC EBeam process (220 nm SOI, oxide cladding) for TE operation at 1550 nm. The free spectral range (FSR) of each MZI will be used to extract the group index n_g of the $500 \times 220\text{ nm}$ strip waveguide. A finite-element mode solver predicts $n_g = 4.184$ at 1550 nm, and 4.146-4.224 across width/thickness fabrication corners.

1. Theory

A Y-branch splits the input equally into two arms of length L_1 and $L_2 = L_1 + \Delta L$, and a second Y-branch recombines them. With propagation constant $\beta = 2\pi n_{\text{eff}}/\lambda$ and power loss α , the transmission is

$$T(\lambda) = \frac{1}{4} \left| e^{-j\beta L_1 - \alpha L_1/2} + e^{-j\beta L_2 - \alpha L_2/2} \right|^2,$$

which for equal losses reduces to $T = \frac{1}{2}[1 + \cos(\beta\Delta L)]$. Adjacent transmission minima are separated by $\text{FSR} = \lambda^2 / (n_g \Delta L)$, where $n_g = n_{\text{eff}} - \lambda \, dn_{\text{eff}}/d\lambda$. Measuring the FSR of MZIs with known ΔL therefore gives n_g directly; using several ΔL values allows a linear fit of $1/\text{FSR}$ versus ΔL that is insensitive to the fixed routing in both arms.

2. Waveguide modelling

The strip waveguide (Si core, $w = 500\text{ nm}$, $t = 220\text{ nm}$, fully clad in SiO_2) was simulated with the finite-element mode solver femwell, including material dispersion of Si (Li, 1980) and SiO_2 (Malitson, 1965). The fundamental TE mode was solved from 1500 to 1600 nm and fitted to the compact model $n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$, $\lambda_0 = 1.55\text{ }\mu\text{m}$, giving $n_1 = 2.4458$, $n_2 = -1.1211\text{ }\mu\text{m}^{-1}$, $n_3 = -0.0553\text{ }\mu\text{m}^{-2}$ and $n_g(1550\text{ nm}) = 4.1836$ (Fig. 1).

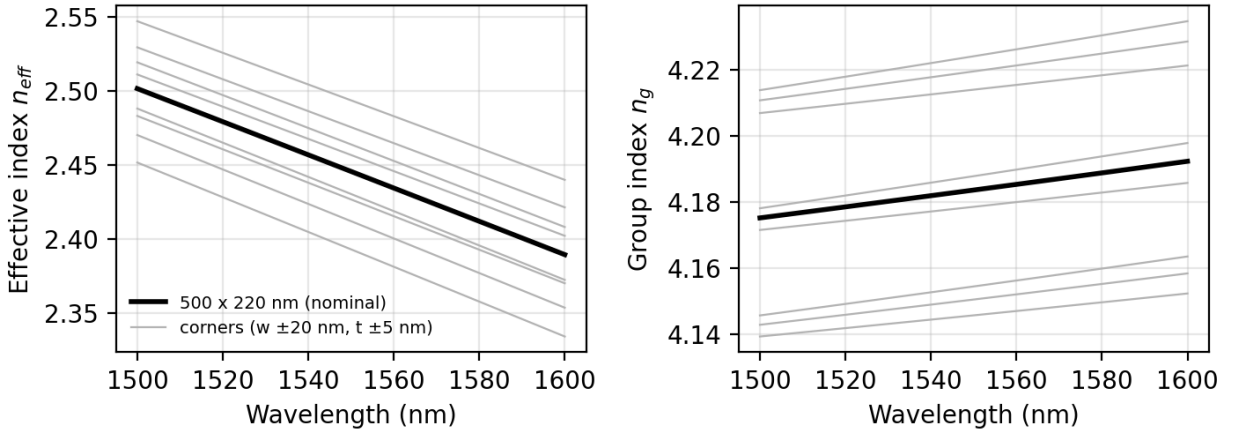


Fig. 1. Simulated effective index (left) and group index (right) of the TE0 mode.

3. MZI design and simulated response

The ΔL values were chosen so that the FSR spans 1-11 nm. The shortest device ($\Delta L = 51\text{ }\mu\text{m}$) still shows several fringes within the grating-coupler pass band, while the longest ($\Delta L = 501\text{ }\mu\text{m}$) gives about 100 samples per fringe at a 10 pm laser step, improving the precision of the extracted n_g . Fig. 2 shows the simulated spectra using the compact model above, ideal Y-branches and 3 dB/cm propagation loss. The common arm length is $L_1 = 128.4\text{ }\mu\text{m}$, extracted from the layout.

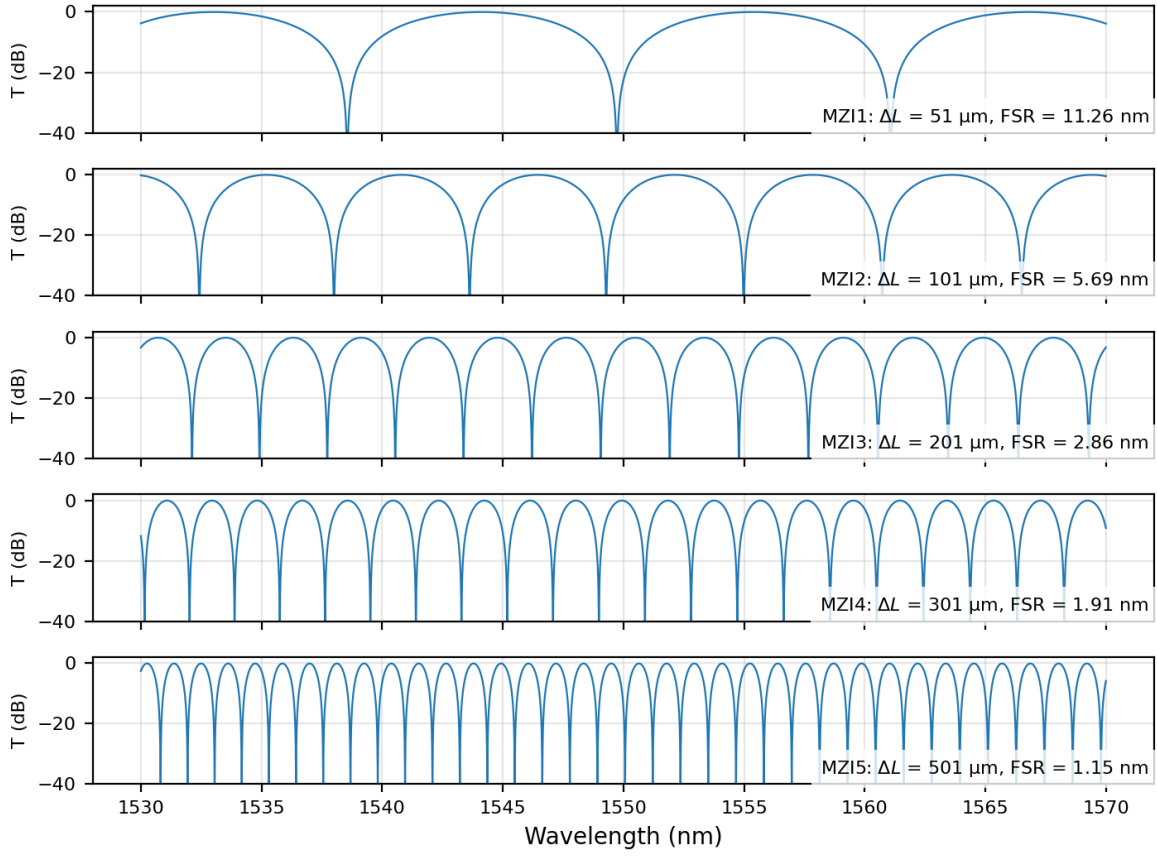


Fig. 2. Simulated MZI transmission near 1550 nm (excluding grating couplers).

4. Fabrication variability

To estimate the effect of fabrication variation, the waveguide was re-simulated at width 480/500/520 nm and thickness 215/220/225 nm. Width dominates: n_g decreases by about 0.035 per +20 nm of width, while thickness changes it by about 0.005 per 5 nm. The resulting FSR range for each device is given in Table 1.

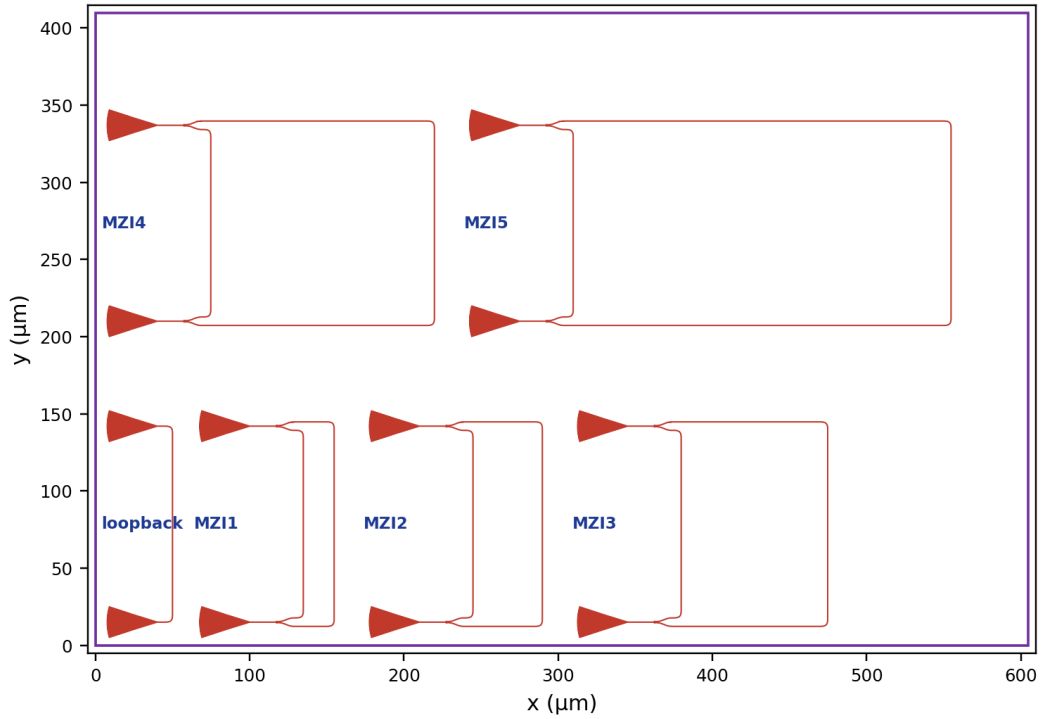
Width (nm)	Thickness (nm)	n_{eff} (1550 nm)	n_g (1550 nm)
480	215	2.3932	4.2140
480	220	2.4123	4.2195
480	225	2.4306	4.2241
500	215	2.4269	4.1785
500	220	2.4458	4.1836
500	225	2.4641	4.1878
520	215	2.4570	4.1457
520	220	2.4757	4.1504
520	225	2.4939	4.1544

Table 1. Devices on the chip and expected FSR at 1550 nm.

opt_in label	Device	ΔL (μm)	FSR nominal (nm)	FSR corners (nm)
opt_in_TE_1550_device_satadrudas_loopback	GC loopback	-	-	-
opt_in_TE_1550_device_satadrudas_MZI1	Y-branch MZI	51	11.26	11.15 - 11.36
opt_in_TE_1550_device_satadrudas_MZI2	Y-branch MZI	101	5.69	5.63 - 5.74
opt_in_TE_1550_device_satadrudas_MZI3	Y-branch MZI	201	2.86	2.83 - 2.88
opt_in_TE_1550_device_satadrudas_MZI4	Y-branch MZI	301	1.91	1.89 - 1.93
opt_in_TE_1550_device_satadrudas_MZI5	Y-branch MZI	501	1.15	1.14 - 1.16

5. Layout

The layout (Fig. 3) was generated with a Python script using KLayout, SiEPIC-Tools and the SiEPIC-EBeam-PDK, within the 605 x 410 μm design area. Each device uses two TE grating couplers (ebeam_gc_te1550) on a 127 μm pitch for the fibre array, two ebeam_y_1550 Y-branches, and Strip TE 1550 nm, $w=500$ nm waveguides with 5 μm bend radius. The upper grating coupler of each pair carries the opt_in label used for automated testing. The loopback (two grating couplers joined by a short waveguide) is used to normalise the grating-coupler response out of the MZI spectra. The layout passes SiEPIC functional verification and the openEBL submission checks with zero errors.

Fig. 3. Layout of EBeam_satadrudas (605 x 410 μm). Bottom row: loopback, MZI1-MZI3; top row: MZI4, MZI5.

6. Test and analysis plan

Each device will be measured with the automated fibre-array setup (TE, 1500-1600 nm sweep). The loopback spectrum is subtracted from each MZI spectrum to remove the grating-coupler envelope. Fringe minima are located to obtain $\text{FSR}(\lambda)$, from which $n_g(\lambda) = \lambda^2 / (\text{FSR} \cdot \Delta L)$ is computed and compared with Fig. 1. Finally, the compact model (n_1, n_2, n_3) is fitted to each measured spectrum, and the spread across the five devices is compared with the corner analysis of Section 4.

References

L. Chrostowski and M. Hochberg, *Silicon Photonics Design*, Cambridge University Press, 2015.

H. H. Li, "Refractive index of silicon and germanium," *J. Phys. Chem. Ref. Data* 9, 561 (1980).

I. H. Malitson, "Interspecimen comparison of the refractive index of fused silica," *JOSA* 55, 1205 (1965).

femwell: finite-element mode solver, <https://github.com/HelgeGehring/femwell>

SiEPIC-EBeam-PDK, https://github.com/SiEPIC/SiEPIC_EBeam_PDK